Léo Saci

PhD student in Machine Learning

Murex AI Research & Université Paris Dauphine – PSL

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GitHub
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Research Experience

Research internship — **Paris, France**
Université Dauphine - PSL — *2024–2025*
COHIRF: a meta-algorithm for clustering.

- Development of a meta-algorithm for clustering.
- Experimental validation.

Research internship — **Toulouse, France**
LAAS-CNRS, Gepetto team — *05/2021–08/2021*
Learning the value function from trajectory optimization using Sobolev descent.

- Reinforcement learning, robotics, optimal control.

Research internship — **Québec, Canada**
Université Laval — *11/2019–08/2020*
Routine bandits: regret minimization in recurring problems.

- Design of a transfer learning algorithm.
- Theoretical analysis and numerical validation.

Research internship — **Gif-sur-Yvette, France**
CEA — *04/2019–09/2019*
Minimization of supervision for incremental deep learning.

- Semi-supervised learning for incremental image classification.

Research internship — **Toulouse, France**
IMT — *04/2018–07/2018*
Method for simulating post-Bayesian log-concave distributions via Langevin diffusion.

- Estimation of posterior means via Langevin Markov diffusion.

Education

PhD in Machine Learning (CIFRE) — **Paris, France**
Murex AI Research / Université Paris Dauphine – PSL — *Since 2025*

Master of Science in Statistical and Financial Engineering (ISF) — **Paris, France**
Université Paris Dauphine – PSL — *2024–2025*

Master of Science in Mathematics, Vision, Learning (MVA) with highest honors — **Cachan, France**
ENS Paris-Saclay — *2018–2019*

Student at ENS Paris-Saclay — **Cachan, France**
Department of Mathematics — *2016–2020*
Admission by competitive exam.

Bachelor's degree in Mathematics, third year, with honors — **Paris, France**
Pierre et Marie Curie University — *2015–2016*

Classes Préparatoires MPSI–MP* — **Paris, France**
Lycée Saint-Louis — *2012–2015*

Publications

[1] Amit Parag, Sébastien Kleff, **Léo Saci**, Nicolas Mansard, and Olivier Stasse. *Value learning from trajectory optimization and Sobolev descent: A step toward reinforcement learning with superlinear convergence properties.* In: 2022 International Conference on Robotics and Automation (ICRA), 2022.

[2] Hassan Saber, **Léo Saci**, Odalric-Ambrym Maillard, and Audrey Durand. *Routine bandits: Minimizing regret on recurring problems.* In: Joint European Conference on Machine Learning and Knowledge Discovery in Databases (ECML

PKDD), 2021.

[**3**] E. Belouadah, A. Popescu, U. Aggarwal, **L. Saci**. *Active class incremental learning for imbalanced datasets.* In: European Conference on Computer Vision (ECCV), 2020.

Teaching

Private mathematics tutoring **Paris, France**
From high school to bachelor's level *Since 09/2016*

Research Interests

Unsupervised learning, probabilistic modeling, reinforcement learning, optimal control, time series, weak supervision.